Development methods

The development and maintenance of applications and systems should be accounted for in the type of governance as well. If the CIO oversees all applications development, then leadership must address that another body will have oversight over the CIO. Sometimes this does not play well with the CIO.

Many organizations have a defined development process, or systems development life cycle, for defining and deploying automated systems. Some aspire to an Agile delivery environment. Others have more traditional "plan and build" approaches. Sadly, many organizations have no formal statement of how to get applications and technology implemented.

Few of them have built any type of consideration for designing and delivering around DG policies and standards. Very often, DG will need to write the enhancements to corporate IT application development when structured information is being governed, or work with compliance to oversee unstructured content. The enhancements take the form of updating traditional project artifacts, and branch out to additional tasks, or new approvals and checkpoints. When unstructured information is subject to DG, we often must modify workflow and document management policy.

Federation

One of the most important concepts affecting the nature and scope of DG is that of "federation." We covered this definition earlier.[2]

For DG, this means defining an entity (the DG program) as a blend of governance capabilities that touch various functions in the organization. The federation of a DG program is a definition of where and how standards will be applied across various layers and segments of an organization. Politically, the United States is a federation, an organization of states with a federal oversight layer. In the United States, some activities of government are central. There is a central military and reserve banking system. Other functions of government operate at the state or local level, such as medical care and law enforcement. A DG program will necessitate the same type of definition of the required layers of governance functionality.

The definition of federation will influence the operations of your DG organization, its processes, and principles. Note in Fig. 4.2, I show a heat map where similar data assets can be tightly governed (in the center or hot zone), or more loosely governed (on the fringe or the cool zone). The solid areas indicate a governed area called "item," where there is tight control of global items, slightly looser control on regional items, and local items are barely governed. The dotted areas point to another subject, "customer." There is still the tight control for centrally used customer content, but the regional and local are treated the same. So, the federated intensity of DG differs by content type.

Scope factors that affect the federated layers and activities are:

- *Enterprise size*—Obviously, huge organizations will need to federate their DG programs, and carefully choose the critical areas where DG adds the most value.
- *Brands*—Organizations with strong brands may want to consider this in their DG scoping exercise. One brand may need a more centrally managed data portfolio than another.
- *Divisions*—One division may be more highly regulated, therefore requiring a different intensity of DG.

[2]The *Webster Dictionary* definition of *federation* offers some insight: *an encompassing political or societal entity formed by uniting smaller or more localized entities: as a: a federal government, b: a union of organizations the act of creating or becoming a federation; especially: the forming of a federal union*.

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- *Countries*—Various nations have different regulations and customs, therefore affecting how you can govern certain types of information.
- *IT portfolio condition*—An organization embarking on a massive overhaul of applications (usually via implementing a large SAP or Oracle enterprise suite) will have definite and specific DG federation requirements. When a DG effort is getting started, it is usually understood (at some intuitive level) what the condition of the application portfolio is. A portfolio undergoing radical updating is a prime target for DG, and the impact on DG scope needs to be clear.
- *Culture and information maturity*—The ability of an organization to use information and data is referred to as its information management maturity, or IMM. The way an organization gets its work done is usually called culture. In combination, the specific IMM and culture of an organization will affect the scope and design of the DG program. For example, an organization that is rigid in its thinking and has a low level of maturity will require more centralized control in its DG program, as well as encounter more significant change management issues.

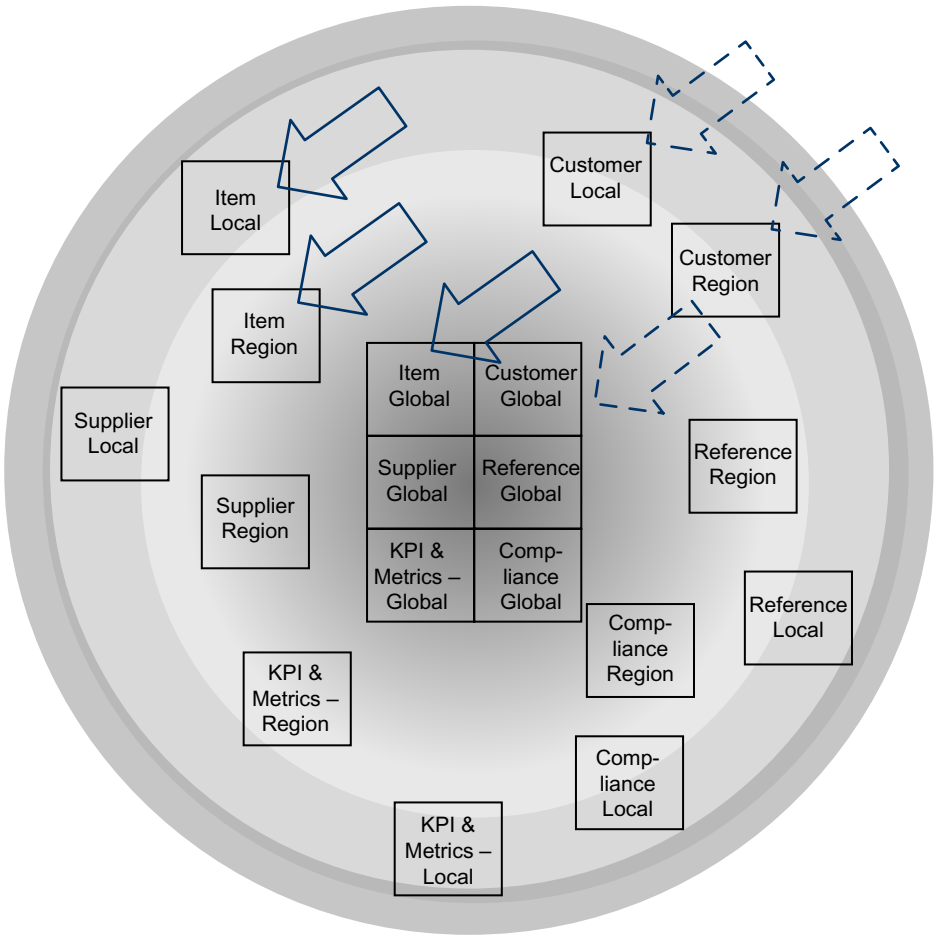


FIG. 4.2

Federation "Heat Map"

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Short case study

Don't fall into the scope trap of identifying the scope of DG with size or market dominance. You need to rationally consider influencing factors we have presented; that is the business model, the assets to be managed, and what type of federation is required. Using Fig. 4.2 again, let's assume it represents a global retail organization, with many brands, but the same distribution and merchandising network.

Business model—the business model is global, with heavy dependence on economy of scale across the supply chain. So, our scope will lean toward the entire organization—we will not be excluding any functions, like merchandising or warehouse.

Content being managed—obviously there is a lot of content in a large organization, but consider the variety—retail is, at its core, pretty simple. You buy stuff from one place and sell it to someone else. The main content is anything used or descriptive of the "stuff" and getting it sold. Be careful—it isn't just the items—what about the people on the sales floor? What about the trucks and trains to move items about? All are integral to the business. So from a scope standpoint we need to consider almost all of the content within this type of enterprise. The key guidance to apply is—the scope of DG is a function of the assets being managed (i.e., the content and information being governed).

Federation—the entire enterprise is in scope, and all content relevant to the business model is in scope. We have not narrowed this down much, have we? When we examine the content (remember we are considering all of it), we see that it stratifies into global, regional, and local. This is significant. If a region or locality can buy items to sell, what is the intensity of DG in those supply chains versus the global ones? We have to consider that local data may not be worth close governance and may be okay with a more relaxed level of intensity.

Bottom line: All content is in scope, but due to size, geography, and markets, we need to consciously identify which specific content is managed centrally, regionally, or locally. The organization would state that DG scope is all content relevant to the business model, but the intensity of DG will vary based on a specifically defined set of federated layers.

Elements of data governance programs

In many ways, a DG program is like any other business program. Many elements of DG make perfect sense to businesspersons when they first consider DG. For some reason, the people on the technology side of the information management and DG equation get dazed and confused. Either way, this section will introduce these basic program elements in the context of DG. We then will review the important, managerial aspects of these elements. Later on, we will get into the specific design and deployment of each of these elements.

DG is simple in its makeup. There are the three cornerstones of any organization—People, Process, and Technology. On top of that, we need to add the newest, permanent cornerstone and aspect of any program—Data. (Do not stop at the People, Process, Technology mantra—you MUST add data).

The People-aspect deals with the various required roles. Process needs to cover an operating model and the capabilities required to operate DG. Technology covers DG-specific technologies. The Data element covers the actual content, metadata, catalogs, all content, data dictionaries, etc.

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Because we are answering the "What does it look like?" question, let's briefly cover the Process element first.

Process—The DG operating model

The DG operating model is made up of two components: the capabilities model, which states WHAT is happening, and the workflow model, which dictates how information flows and parties interact within the program, and how decisions are made.

Essential capabilities

A capabilities approach has become more popular for addressing DG functionality over the years. It is familiar. The basic approach to standing up DG is handled like standing up any other new business capability or upgrading current capabilities to include better data behaviors. We use the term capability to describe the "what" has to happen in DG. Because it is programmatic, DG introduces new capabilities to the organization.

Capabilities are also much more stable than processes or functions. Since DG is a long-term alteration in data operations, it makes sense to be more stable. In a world where agility is important, capability-based design allows iterations more easily, since your target is putting in stable capabilities vs a specific process tied to a policy.

Initially these functions will appear to be embedded in the DG "area" but over time they need to evolve into day-to-day activities within all areas. Using the word "process" would immediately imply a "where," and that is an operational level of detail that evolves. Within the capabilities are functions and activities that will be visible in the DG management framework as stand-alone processes as well as the day-to-day activities that carry out governance. There is no need to design your DG functional model from scratch; there is a list later in the book. However, recognizing that there will be a formal set of functional requirements (to be manifested as processes) and that they will be executed all the time is a key element to the success of DG. Capabilities perform two roles. First, they point out what someone actually must do. Second, reviewing the functions required for your organization usually aids in determining which areas or individuals would bear accountability and responsibility.

The DG area will need to consider other business areas where there will be interaction and collaboration, such as:

- Human resources
- Compliance and/or legal
- Risk management
- Large-scale integration projects, such as enterprise resource planning

The bottom line for this element of DG is that you need to formally consider and build the DG processes and functions. They are not instinctive.

In the first edition I called out FUNCTIONS to indicate the WHAT. However, "Capabilities" fits better. Lastly, this is the language of enterprise architecture, and over the past years I have noticed it is easier to get DG engaged with enterprise architects, as well as the usual constituents. Table 4.1 shows a very high-level representation of DG capabilities.

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Table 4.1 High-level data governance capabilities

High level data governance capabilities
Data and governance strategy
Data governance requirements and design
Data governance frameworks
Supporting technology
Data governance operation
Data governance engagement
Data governance measurements
Technology operation
Communication
Training
Data governance services
Sustaining design
Sustaining activity management
Sustaining program operation

Work flow

Like any other activity within a company or government entity, there needs to be a formal statement of work flow. Please note that this is no hint to design an organization chart. Rarely does DG activity become a stand-alone area (i.e., there is rarely a DG "department").

Work will flow within DG like other forms of governance. In lieu of an organization chart, consider DG structure to be more of a matrix. This allows incredible flexibility (Table 4.2).

Table 4.2 Data governance as work flow

	Management	Operations
Strategic activity	Set strategy, align to business, provide principles, ultimate issue resolution	Specify controls, metrics
Tactical activity	Identify issues, offer coaching, support application of policy	Provide DG services, operate supporting technology, provide training and communications

Perhaps the best way to see DG capabilities in action is to look at a common artifact from an active DG program—a meeting agenda for the main DG council. The council represents a group that gets the governing done (Fig. 4.3).

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Council Agenda Planning					March 1, 20nn
Month:	**March 7th**		**CURRENT AGENDA**		
Presenter	Topic		Item	Duration	Action required
	Orientation		Charter ratification discussion	10	Understand role of the Data Governance Council (DGC)
			Principles and policy introduction	20	Understand role of principles and policies
	DG Value		Recommended DG metrics	5	Understand role of metrics to measure DG and review proposed measures for DG
			DG scorecard introduction	10	Review current DG scorecard
	DG Compliance		Principles and policy roll-out	15	Determine / approve principle / policy roll out
			Status of classification policy and efforts		Awareness of effort and business impacts
	DG Process		DGC meeting protocol	10	Define meeting protocol
	Issues		Custodian training	20	Trial issue—Review and recommend resolution
				90	
Month:	**April 3rd**		**FUTURE AGENDA**		
Presenter	Topic		Item	Duration	Action required
	DG Process		DGC meeting protocol, rules of order	10	Define meeting protocol
	Orientation		Education—Role of DG	10	Undertand role of principles and policies
			Demystifying policy presentation	10	Undertand policy hierarchy
			Sample DG scenarios currently underway	15	Undertand what is going on
	DG Roll-out		Crawl, walk, run approach	15	Review and adjust/approve DGC “road map”
	DG Value		Short term objectives—walk the talk	15	Leave with consistent identity and message
				65	

FIG. 4.3

Sample data governance agenda

Helpful hint

Remember that the ultimate goal of DG is to disappear as a stand-alone program. It becomes part of the fabric of business, like financial controls. That is why the DG “department” is really a monitoring structure—much like an audit committee. To be more specific, DG may not really

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disappear, but it will be very thin. There will always be the need to resolve issues. But like other types of corporate governance, these events become accepted as normal activities, not special programs.

There may be a thin department with participants that roll in and roll out. Some highly regulated organizations may want to have a separate DG department only if it cannot be fit into the compliance areas. This is one of those areas we promised to point out—we differ from many of our peers. Given the long-term nature of DG (i.e., it's not a lot different than financial controls or well-known policies), there is little need for a full-time overhead structure. In our opinion, that perpetuates the labeling of data management and governance as programs that can be terminated, as opposed to the permanent behavior change it really represents.

A specific time frame for DG to reach the "transparent" stage is hard to define, as it will vary based on scope and organization. The degree of transparency will be tied to your progress up the IMM curve, so whatever the timing is of your IMM progress is most likely your timing for DG to fade into the business fabric. If you are using a five-stage maturity model to measure DG effectiveness, then whenever you hit stage 4 or 5, your DG program should be part of your everyday activity. This may take a very long time. Think about it more as a goal rather than a requirement.

Again, think about financial controls. Few organizations talk about the financial governance program and whether it should be justified to continue or be terminated. That would be unheard of. This perception is what you are striving for with DG.

Regardless of the size of the organization or the complexity of DG, it is key to remember the DG organization is not there to do information management. There will be DG functions and processes to manifest the capabilities. (There is a complete list of sample functions in the appendices.)

Principles

We touched on principles earlier by way of a definition. In summary, they are general adopted statements of philosophy that guide conduct and application of data management and governance capabilities. Principles are more than just a term to be understood, however. Principles are crucial elements in DG. One client told me they justified the entire program because with principles in place, there were fewer meetings.

Principles will succeed where a batch of rules and policies will not. They are foundational. One explanation we use when confronted with resistance to developing principles is to draw an analogy to the "Bill of Rights," the first 10 amendments to the United States Constitution. It's easy to see the historical significance of the application of these principles to United States history. It is the same for data principles (maybe a bit less historical).

As you deploy DG, you will need to revisit and repeat your enterprise-level principles. Not revise but repeat. Since they are foundational and represent beliefs, repetition will be necessary. Table 4.3 lists some sample principles we have collected. (Please note that I occasionally refer to an organization named "Farfel." This is the name of a fictional company used as a sample case study and will appear again in later chapters.)

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Table 4.3 Sample principles

Principle name	Principle description
Master principle	Enterprise data will be governed by a formal organization, with appropriate authority and accountability to define and establish how information, data, and content is managed.
Federation	FARFEL will have enterprise standards and guidelines for all metrics, content, data structures, codes, values, and data naming.
Data efficiency	Data, information, and content needs to be available at the right time, at the right place, and in the right format to authorized users/consumers, at an efficient cost.
Business alignment	Information management solutions will maintain business alignment, and will only be in response to business needs vs business area requests.
Information quality	Enterprise data will be managed and measured for quality. There will be parties that are accountable for overall integrity and quality of enterprise data and content.
Risk management	Appropriate due diligence will be conducted to ensure data complies with all applicable statutes and regulations.
Share and collaborate	Enterprise data is a shared resource across the enterprise. Data is not a resource that can be "owned" by specific business areas.

Policies

Another element we previously defined is policies. Policies are formally defined processes with strength of support—that is, they are a codification of a principle. They give it "teeth." Policies include standards—one area where IT personnel will be very intense as DG becomes real. Most likely, you already have most of your DG policies floating around in the form of a disconnected IT, data, or compliance policy. And, like most places, the policy sits happily in its notebook while life goes on and the policy is disregarded. The marriage of principle and policy prevents this in the DG program.

Metrics

You cannot manage what you do not measure. Over time, your DG program will need to evolve a means to monitor its own effectiveness. Without it, the DG program will certainly fade away. At the outset, the metrics will be hard to collect. After all, you have not been managing data very well, so there is no infrastructure to install a metric. Eventually, the metrics will evolve from simple surveys and counts to true monitoring of activity. By way of explanation, here is a brief list of common metrics:

- *DG Stewardship Progress*—Report on counts of individuals trained on DG, counts of specific projects governed, and a count of issues elevated and/or resolved.
- *DG Stewardship Effectiveness*—Alternatively to progress, an effective metric can be based on counts and resolution of issues submitted to DG bodies.
- *Data Quality*—Data profiling results calculated into a DQ index that represent an average of all of the data-quality profiling measures.
- *DG Value*—We will dive into the business case and business value more in the next chapter, but you can never go wrong with tying the application of DG and data management to business success. Quantifiable and intangible benefits resulting from successful efforts that were governed, or through use of governed and well-managed data, should always be reported.

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The people element—Roles and responsibilities

The official designation of accountability and responsibility are key factors to the survival of DG. Most important to new DG programs is the concept of accountability for data. This is most likely a very new role. To be clear, it will seem very new and different to hold someone accountable for data quality—especially when accountability means a direct effect on bonuses or promotions. There will also be a perception that the DG program is rather powerful or bold to be making these designations. Assigning responsibility will also be an important activity. In many organizations, the responsible parties have a formal role as designated "stewards" or "custodians." Other implementations of DG may place everyone under a label of a steward, and the responsible parties will be direct supervisors.

Many organization view DG as a door through which to introduce new roles. The new roles tend to focus around a concept of placing an individual in charge of data assets, usually at a level above and outside of information technology. This "top data job" places formal accountability for management of data in one place. The Chief Data Officer, or CDO, is a manifestation of this job. Some CDO or equivalent titles will report to the CEO. Others may report to, or act as, an office in charge of data monetization or analytics. Either way, these organization find that data assets are important enough to warrant a new line of authority. In my practice I look for opportunities to get a top data job identified and placed. It helps a great deal.

Data

Governing data means understanding your data. That means the operation of DG will help oversee managing data wherever it is, and however it moves around. When planning and operating DG, this means the program needs to be aware of the data landscape.

The data landscape is your inventory of data—what do you have, where is it, who uses it, where does it come from. Often a lineage or provenance capability is also required as part of managing the data landscape.

The data assets of the organization, represented by the landscape, are the operational targets of DG, whereby it oversees the usage, movement, interfacing, and integrity of the data.

Most issues in DG will come from application of governance to a data area that is confused or resistant to change. Hence, Data needs to be a distinct focus area alongside People, Process, and Technology.

Technology and tools

The last element that requires high-level consideration is Technology. As of the writing of this edition, the market for pure DG technology is evolving rapidly. Specific tools for data glossary management, DG workflow, data discovery, and data quality/governance integration are being brought to market. These are on top of the traditional technologies in support of DG, such as SharePoint, Word, and Excel, as well as adapting tools from other disciplines, like data model or data dictionary tools. Specialty tools are evolving and, in general, you will want to consider the following capabilities, but Chapter 15 will cover the application of tools in more detail.

One aspect of tools to understand at this point is that you should not feel compelled to buy DG tools just because you are doing DG. By definition, a tool exists to improve something you are already doing. If you are not doing formal DG yet, or if you are doing it poorly, then casting about for a tool to help you deploy DG is a waste of time. This flies in the face of typical IT philosophy, where the tool is usually acquired first. It's a notoriously silly thing to do. Our work always has us putting the brakes on

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a tool selection project: It's easy to buy a tool and install it. But most of the time we witness new tools for data management sitting unused or poorly deployed, because no one has mastered the process the tool is supporting.

As you roll out DG and begin to understand the various aspects of your particular program, you will know immediately where you need a tool to "grease the skids." Some features of DG tools to consider are:

- Principle and policy administration
- Business rules and standards administration
- Organization management
- Work flow for issues and audits
- Data discovery
- Taxonomy or ontology management
- Data dictionary
- Enterprise search
- Document management
- Metrics scorecard—data gathering, synthesis, and presentation
- Interfaces to other workflows and methodologies
- Training and collaboration facilities

The critical success factors for data governance

Because DG is a business program, we need to point out the critical success factors (CSFs) early on in this book. Frankly, if one or more of the CSFs presented next are totally unrealistic for your organization, you need to reconsider launching a formal DG program as an approach to improving data asset management. Or at least, you should call it something else.

1. DG is mandatory for the successful implementation of any project or initiative that uses information. Any project requiring reports, business intelligence, cleaning of data, or development of a "single source of truth" requires DG to be sustainable and successful.
2. DG must show value explicitly. This means you cannot do DG in a vacuum. Something has to be governed, even if it is data quality and you implement DG as a means to improve data quality. Countless IT shops developed models, standards, and policies in the 1980s and 1990s, and then went looking for a project to spring them on. You need to show benefit, and that means tying the DG effort to a visible initiative.
3. You must manage organizational change. At the risk of being repetitive, you are doing DG because you are NOT doing something correctly. Therefore, something needs to be changed. We have dealt with numerous organizations who wanted all of their data fixed but did not want to change their views or the behaviors or processes that created the mess. So, you will need to orient, train, educate, communicate, hold hands, encourage, and offer incentives. Then repeat it all again.
4. DG must be viewed as an enterprise effort. You can implement it in segments, but it must always have an enterprise perspective. Otherwise, you will end up with conflicting standards and accountabilities.

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